Chapter 8

Operators on Complex Vector Spaces

Contents

8A Generalized Eigenvectors and Nilpotent Operators

1 Suppose $T \in \mathcal{L}(V)$. Prove that if $\dim \operatorname{null} T^4 = 8$ and $\dim \operatorname{null} T^6 = 9$, then $\dim \operatorname{null} T^m = 9$ for all integers $m \geq 5$.

Solution:

By Theorem 8.1, dimension of $\operatorname{null} T^5$ can be either 8 or 9.

If $\dim \operatorname{null} T^5 = 8$, then by Theorem 8.2, $\dim \operatorname{null} T^6 = 8$, which contradicts the initial assumptions. Hence $\dim \operatorname{null} T^5 = 9$. Now Theorem 8.2 implies that $\dim \operatorname{null} T^m = 9$ for all integers $m \geq 5$. $\square$

2 Suppose $T \in \mathcal{L}(V)$, m is a positive integer, $v \in V$, and $T^{m-1}v \neq 0$ but $T^m v = 0$. Prove that $v, Tv, T^2v, \dots, T^{m-1}v$ is linearly independent.

Solution:

Suppose that the list $v, Tv, T^2v, \dots, T^{m-1}v$ is linearly dependent. That means there exist nonzero $a_0, a_1, \dots, a_{m-1} \in \mathbb{F}$ such that

$$a_0 v + a_1 T v + a_2 T^2 v + \dots + a_{m-2} T^{m-2} + a_{m-1} T^{m-1} = 0. \tag{8.1}$$

Apply T to both sides, which results in (using $T^m v = 0$):

$$a_0 T v + a_1 T^2 v + \dots + a_{m-2} T^{m-1} = 0.$$

Let $k \in \{0, \dots, m-1\}$ be the smallest integer such that $a_k \neq 0$. If $k = m-1$, then we have a contradiction to our assumption right away. Suppose $k \neq m-1$. Then apply operator T as was shown above $(m-1-k)$ times to eq. 8.1. That results in

$$a_k T^{m-1} v = 0.$$

This implies $T^{m-1}v = 0$, contradicting the initial assumption.

Thus, $v, Tv, T^2v, \dots, T^{m-1}v$ is linearly independent, as desired. $\square$

3 Suppose $T \in \mathcal{L}(V)$. Prove that

$$V = \operatorname{null} T \oplus \operatorname{range} T \iff \operatorname{null} T^2 = \operatorname{null} T.$$

Solution:

First suppose that $V = \operatorname{null} T \oplus \operatorname{range} T$. Thus, every $v \in V$ can be uniquely represented as $v = u + w$, where $u \in \operatorname{null} T$ and $w \in \operatorname{range} T$.

Note that $\operatorname{null} T \subseteq \operatorname{null} T^2$, by Theorem 8.1. We need to show the inclusion in the other direction. Let $v \in \operatorname{null} T^2$. As stated earlier, we can write $v = u + w$. Then we have:

$$0 = T^2 v = T(Tu + Tw) = T(0 + Tw) = T(Tw). \tag{8.2}$$

If $w = 0$, then we of course have $v \in \operatorname{null} T$. If $w \neq 0$, then we have $Tw \in \operatorname{range} T$ and $Tw \neq 0$, otherwise it would have been an element of $\operatorname{null} T$. Similarly, we have $T(Tw) \in \operatorname{range} T$ and $T(Tw) \neq 0$, as $Tw \notin \operatorname{null} T$. This contradicts eq. 8.2, implying that w must equal zero.

Hence, $\operatorname{null} T^2 \subseteq \operatorname{null} T$, which completes this part of the proof.

Now suppose $\operatorname{null} T^2 = \operatorname{null} T$. Theorem 8.2 implies that $\operatorname{null} T^{\dim V} = \operatorname{null} T$.

Using Theorem 8.4, we can write

$$V = \operatorname{null} T^{\dim V} \oplus \operatorname{range} T^{\dim V} = \operatorname{null} T \oplus \operatorname{range} T^{\dim V}. \tag{8.3}$$

Note that $\operatorname{range} T^{\dim V} \subseteq \operatorname{range} T$. Also we have $\dim \operatorname{range} T^{\dim V} = \dim V - \dim \operatorname{null} T$ from eq. 8.3, and $\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T$ from the Fundamental Theorem of Linear Maps. This implies (by Theorem 2.39) that $\operatorname{range} T^{\dim V} = \operatorname{range} T$. Continuing with eq. 8.3, we get

$$V = \operatorname{null} T \oplus \operatorname{range} T.$$

That finishes the proof in the other direction, establishing the desired result. □

4 Suppose $T \in \mathcal{L}(V)$, $\lambda \in \mathbb{F}$, and m is a positive integer such that the minimal polynomial of T is a polynomial multiple of $(z - \lambda)^m$. Prove that

$$\dim \operatorname{null} (T - \lambda I)^m \geq m.$$

Solution:

Denote $T - \lambda I$ by S. Suppose that $\dim \operatorname{null} S^m < m$. Let $\mu = \dim \operatorname{null} S$ ($\mu < m$).

Now the Theorem 8.3 implies $\operatorname{null} S^m = \operatorname{null} S^\mu$. Hence for every $u \in \operatorname{null} S^m$ we have $S^\mu u = 0$.

Let p denote the minimal polynomial of T, and q be a polynomial such that $p = (z-\lambda)^m \cdot q$. Consider $v \in V$. If $v \notin \operatorname{null}(T-\lambda I)^m$, we still have $p(T)v = 0$, which implies $q(T)v \in \operatorname{null}(T-\lambda I)^m$. By our considerations in the previous paragraph we thus have $(T-\lambda I)^\mu(q(T)v) = 0$, which implies $(T-\lambda I)^\mu q(T)v = 0$ for every $v \in V$. This contradicts minimality of $p(z)$. Therefore, $\dim \operatorname{null}(T-\lambda I)^m$ cannot be less than m. $\square$

5 Suppose $T \in \mathcal{L}(V)$ and m is a positive integer. Prove that

$$\dim \operatorname{null} T^m \leq m \dim \operatorname{null} T.$$

Solution:

If $m = 1$, then the claim is obviously true.

Now suppose that for some positive integer k it is true that $\dim \operatorname{null} T^k \leq k \dim \operatorname{null} T$. Consider $m = k+1$. We have

$$\begin{aligned}\dim \operatorname{null} T^{k+1} &= \dim \operatorname{null} T^k T \leq \dim \operatorname{null} T^k + \dim \operatorname{null} T \\ &\leq k \dim \operatorname{null} T + \dim \operatorname{null} T \\ &= (k+1) \dim \operatorname{null} T,\end{aligned}$$

where the first inequality comes from *Problem 3B.22*, and the last inequality comes from the induction hypothesis.

Thus, $\dim \operatorname{null} T^m \leq m \dim \operatorname{null} T$ is true for every positive integer m by induction. $\square$

6 Suppose $T \in \mathcal{L}(V)$. Show that

$$V = \operatorname{range} T^0 \supseteq \operatorname{range} T^1 \supseteq \cdots \supseteq \operatorname{range} T^k \supseteq \operatorname{range} T^{k+1} \supseteq \cdots.$$

Solution:

T^0 is just the identity operator, hence $\operatorname{range} T^0 = V$. Then, by Theorem 3.18, $\operatorname{range} T$ is a subspace of V, hence $\operatorname{range} T \subseteq \operatorname{range} T^0$.

Suppose k is some positive integer. Then Theorem 5.18 states that $\operatorname{range} T^k$ is invariant under T. In other words, if $T^{k+1}v \in \operatorname{range} T^{k+1}$, it can be written as $T(T^k v)$, and it is an element of $\operatorname{range} T^k$. That shows $\operatorname{range} T^{k+1} \subseteq \operatorname{range} T^k$ and completes the proof. $\square$

7 Suppose $T \in \mathcal{L}(V)$ and m is a nonnegative integer such that

$$\operatorname{range} T^m = \operatorname{range} T^{m+1}.$$

Prove that $\operatorname{range} T^k = \operatorname{range} T^m$ for all $k > m$.

Solution:

Suppose it is true for some $k > m$. Consider $k+1$. Let $v \in V$, then

$$T^{k+1}v = T(T^k v).$$

Denote $T^k v$ by u. Note that $u \in \operatorname{range} T^k$, hence $u \in \operatorname{range} T^m$. Then $T^{k+1}v = Tu \in \operatorname{range} T^{m+1} = \operatorname{range} T^m$. This implies $\operatorname{range} T^k = \operatorname{range} T^m$ for all $k > m$, as desired. $\square$

8 Suppose $T \in \mathcal{L}(V)$. Prove that

$$\operatorname{range} T^{\dim V} = \operatorname{range} T^{\dim V+1} = \operatorname{range} T^{\dim V+2} = \cdots.$$

Solution:

We only need to show $\operatorname{range} T^{\dim V} = \operatorname{range} T^{\dim V+1}$, the rest follows from *Problem 8A.7*.

Suppose this is not true. Then the results of *Problem 8A.6* and *7* imply

$$V = \operatorname{range} T^0 \supsetneq \operatorname{range} T^1 \supsetneq \cdots \supsetneq \operatorname{range} T^{\dim V} \supsetneq \operatorname{range} T^{\dim V+1}.$$

At each step the dimension of the subspace decreases by at least one. Thus, $\dim \operatorname{range} T^{\dim V+1} < 0$, which is impossible. Thus, the claim of the problem is true. $\square$

9 Suppose $T \in \mathcal{L}(V)$ and m is a nonnegative integer. Prove that

$$\operatorname{null} T^m = \operatorname{null} T^{m+1} \iff \operatorname{range} T^m = \operatorname{range} T^{m+1}.$$

Solution:

If $m \geq \dim V$, then the equivalence is trivially true. Later we suppose $m < \dim V$.

First suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$. Theorem 8.2 implies that $\operatorname{null} T^{\dim V} = \operatorname{null} T^m$.

Using Theorem 8.4, we can write

$$V = \operatorname{null} T^{\dim V} \oplus \operatorname{range} T^{\dim V} = \operatorname{null} T^m \oplus \operatorname{range} T^{\dim V}. \tag{8.4}$$

By the result of *Problem 8A.6*, $\operatorname{range} T^{\dim V} \subseteq \operatorname{range} T^m$.

Now we have $\dim \operatorname{range} T^{\dim V} = \dim V - \dim \operatorname{null} T^m$ from eq. 8.4, and $\dim \operatorname{range} T^m = \dim V - \dim \operatorname{null} T^m$ from the Fundamental Theorem of Linear Maps. That implies $\dim \operatorname{range} T^m = \dim \operatorname{range} T^{\dim V}$, hence $\operatorname{range} T^m = \operatorname{range} T^{\dim V}$. Now, by the result of *Problem 8A.6*, $\operatorname{range} T^m = \operatorname{range} T^{m+1}$, completing the first part of the proof.

Now suppose $\operatorname{range} T^m = \operatorname{range} T^{m+1}$. The *Problem 8A.7* then implies $\operatorname{range} T^m = \operatorname{range} T^{\dim V}$.

Using Theorem 8.4, we can write

$$V = \operatorname{null} T^{\dim V} \oplus \operatorname{range} T^{\dim V} = \operatorname{null} T^{\dim V} \oplus \operatorname{range} T^m. \tag{8.5}$$

By Theorem 8.1, $\operatorname{null} T^m \subseteq \operatorname{null} T^{\dim V}$.

Now we have $\dim \operatorname{null} T^{\dim V} = \dim V - \dim \operatorname{range} T^m$ from eq. 8.5, and $\dim \operatorname{null} T^m = \dim V - \dim \operatorname{range} T^m$ from the Fundamental Theorem of Linear Maps. That implies $\dim \operatorname{null} T^m = \dim \operatorname{null} T^{\dim V}$, hence $\operatorname{null} T^m = \operatorname{null} T^{\dim V}$. Now Theorem 8.1 implies $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, completing the proof. $\square$

10 Define $T \in \mathcal{L}(\mathbb{C}^2)$ by $T(w, z) = (z, 0)$. Find all generalized eigenvectors of T.

Solution:

Operator T has the only eigenvalue, 0. Note that $T^2(w, z) = (0, 0)$, hence every $v \in \mathbb{C}^2$ is a generalized eigenvector corresponding the the eigenvalue 0.

11 Suppose that $T \in \mathcal{L}(V)$. Prove that there is a basis of V consisting of generalized eigenvectors of T if and only if the minimal polynomial of T equals $(z - \lambda_1) \ldots (z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

Solution:

If $\mathbb{F} = \mathbb{C}$, then Theorem 5.47 immediately implies the result.

Let $\mathbb{F} = \mathbb{R}$.

First suppose there is a basis of V consisting of generalized eigenvectors of T: $v_1, \ldots, v_n$. Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of T corresponding to the generalized eigenvectors (possibly, repeated). By Theorem 8.3, for each generalized eigenvector

$$(T - \lambda_j I)^{\dim V} v_j = 0.$$

Construct the polynomial $q(z) = (z - \lambda_1)^{\dim V} \ldots (z - \lambda_n)^{\dim V}$. We have

$$q(T)v_j = 0,$$

for each generalized eigenvector v_j, which follows from property 5.17b. Thus, $q(T) = 0$. Now Theorem 5.29 implies that $q(z)$ is the polynomial multiple of the minimal polynomial of T. Hence, the minimal polynomial of T has the form $(z-\lambda_1)\dots(z-\lambda_m)$ for some $\lambda_1,\dots,\lambda_m \in \mathbb{F}$.

Now suppose the minimal polynomial of T equals $(z-\lambda_1)\dots(z-\lambda_m)$ for some $\lambda_1,\dots,\lambda_m \in \mathbb{F}$. Hence, the eigenvalues of T exist and the proof of Theorem 8.9 applies here. Therefore, the basis of V consisting of generalized eigenvectors of T exists. $\square$

12 Suppose $T \in \mathcal{L}(V)$ is such that every vector in V is a generalized eigenvector of T. Prove that there exists $\lambda \in \mathbb{F}$ such that $T - \lambda I$ is nilpotent.

Solution:

First we show that any $v, w \in V$ correspond to the same eigenvalue.

Suppose α, β, and γ are eigenvalues corresponding to generalized eigenvectors v, w, and $v+w$. As v, w, $v+w$ is a linearly dependent list, at least two of the eigenvalues must be the same. If $\alpha = \beta$, we can stop here. If not, then suppose without loss of generality that $\alpha = \gamma$. Then we have $(T-\alpha I)^{\dim V} v = 0$ and $(T-\alpha I)^{\dim V}(v+w) = 0$, but also

$$(T-\alpha I)^{\dim V}(v+w) = (T-\alpha I)^{\dim V} v + (T-\alpha I)^{\dim V} w = (T-\alpha I)^{\dim V} w.$$

Hence, the generalized eigenvector w also corresponds to the eigenvalue α.

Thus, every pair of vectors in V must correspond to the same eigenvalue. By Theorem 8.11, each generalized eigenvector cannot correspond to more than one eigenvalue, hence there is a common eigenvalue for all vectors in V, say λ. Thus, $T - \lambda I$ is a nilpotent operator. $\square$

13 Suppose $S, T \in \mathcal{L}(V)$ and ST is nilpotent. Prove that TS is nilpotent.

Let k be an integer such that $(ST)^k = 0$. Now consider $(TS)^{k+1}$. Let $v \in V$, then

$$(TS)^{k+1} v = T(ST)^k(Sv) = T(0) = 0.$$

Hence, TS is also a nilpotent operator. $\square$

14 Suppose $T \in \mathcal{L}(V)$ is nilpotent and $T \neq 0$. Prove that T is not diagonalizable.

Solution:

Suppose T were diagonalizable. Then Theorem 8.17 implies that 0 is the only eigenvalue of T. Now, by definition of a diagonalizable operator, its matrix with respect to the corresponding basis has all entries equal to zero except for diagonal entries, which are equal to eigenvalues of T. As 0 is the only eigenvalue of T, all diagonal entries are zero, which implies $T = 0$, contradicting the initial condition $T \neq 0$.

Therefore, T is not diagonalizable. $\square$

15 Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Prove that T is diagonalizable if and only if every generalized eigenvector of T is an eigenvector of T.

Solution:

If T is diagonalizable, than there is a basis of V consisting of eigenvectors of T (Theorem 5.55). Each eigenvector is also a generalized eigenvector. The eigenvectors of the basis are the only linearly independent generalized eigenvectors, hence all generalized eigenvectors are eigenvectors of T.

Now suppose that every generalized eigenvector of T is an eigenvector of T. Theorem 8.9 states that there exists a basis of V, consisting of generalized eigenvectors of T. This same basis is a basis consisting of eigenvectors of T. Hence, Theorem 5.55 implies that T is diagonalizable. $\square$

16 (a) Give an example of nilpotent operators S, T on the same vector space such that neither $S + T$ nor ST is nilpotent.

(b) Suppose $S, T \in \mathcal{L}(V)$ are nilpotent and $ST = TS$. Prove that $S + T$ and ST are nilpotent.

Solution:

Let T and S be defined by the following matrices with respect to the standard basis of $\mathbb{F}^2$:

$$T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

First, we have (using matrix multiplication):

$$ST = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

From the matrix of ST we clearly see that one of the eigenvalues of ST is 1, hence ST is not nilpotent (see Theorem 8.17).

Next,

$$S + T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

From here we can work out the eigenvalues of $S + T$, which are 1 and -1. Thus, $S + T$ is also not nilpotent.

(b) Let $k, m \in \mathbb{N}$ be the least such that $S^k = 0$ and $T^m = 0$. Without loss of generality suppose $k > m$.

First, consider ST:

$$(ST)^k = ST \ldots ST = S^k T^k = 0,$$

where we used commutativity in the second equality, and nilpotence of S and T in the last. Thus, ST is a nilpotent operator.

Second, consider $S+T$. Let $n = 2k$, so

$$(S+T)^n = \sum_{j=0}^{n} C_n^j S^j T^{n-j} = 0,$$

where C_n^j are binomial coefficients. Here we can write the first equality because S and T commute. The second equality comes from the fact, that every term has either S, or T to a power greater or equal to k. As $S^j = 0$ and $T^j = 0$ for $j \geq k$, every term equals zero.

Thus, $S+T$ is also a nilpotent operator. □

17 Suppose $T \in \mathcal{L}(V)$ is nilpotent and m is a positive integer such that $T^m = 0$.

(a) Prove that $I - T$ is invertible and that $(I-T)^{-1} = I + T + \cdots + T^{m-1}$.

(b) Explain how you would guess the formula above.

Solution:

One is not an eigenvalue of T (Theorem 8.17), hence $T - I$ is invertible (Theorem 5.7).

Apply $(I-T)$ to the right side of the claimed equality:

$$\begin{aligned}(I-T)(I+T+\cdots+T^{m-1}) &= I + T + \cdots + T^{m-1} - T - T^2 - \cdots - T^{m-1} - T^m \\ &= I - T^m \\ &= I,\end{aligned}$$

where we cancelled all similar terms in the second equality, and used $T^m = 0$ in the third.

As both factors are polynomials of T, they commute with each other. Hence, by definition of inverse, we have

$$(I-T)^{-1} = I + T + \cdots + T^{m-1}. \qquad \square$$

(b) This formula can be guessed from the Taylor expansion for $1/(1-x)$.

18 Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that $T^{1+\dim \operatorname{range} T} = 0$.

Solution:

Note that $\operatorname{range} T$ is invariant under T^k for any integer k. Thus, let us focus on $T|_{\operatorname{range} T}$.

As T is nilpotent, so is $T|_{\operatorname{range} T}$. Applying Theorem 8.16 to it, we get $(T|_{\operatorname{range} T})^{\dim \operatorname{range} T} = 0$. In other words, $T^{\dim \operatorname{range} T} u = 0$ for any $u \in \operatorname{range} T$.

Now let $v \in V$, then $Tv \in \operatorname{range} T$. Hence,

$$T^{1+\dim \operatorname{range} T} v = T^{\dim \operatorname{range} T}(Tv) = 0,$$

which implies $T^{1+\dim \operatorname{range} T} = 0$. $\square$

19 Suppose $T \in \mathcal{L}(V)$ is not nilpotent. Show that

$$V = \operatorname{null} T^{\dim V - 1} \oplus \operatorname{range} T^{\dim V - 1}.$$

Solution:

We start from Theorem 8.4:

$$V = \operatorname{null} T^{\dim V} \oplus \operatorname{range} T^{\dim V}. \tag{8.6}$$

T is not nilpotent, hence $\dim \operatorname{null} T^{\dim V} < \dim V$. Now Theorems 8.1 and 8.2 imply that $\operatorname{null} T^{\dim V - 1} = \operatorname{null} T^{\dim V}$, otherwise $\dim \operatorname{null} T^{\dim V}$ would have been equal to $\dim V$.

Result of *Problem 8A.9* shows that $\operatorname{null} T^{\dim V - 1} = \operatorname{null} T^{\dim V}$ implies $\operatorname{range} T^{\dim V - 1} = \operatorname{range} T^{\dim V}$. Thus, we can rewrite eq. 8.6 as

$$V = \operatorname{null} T^{\dim V - 1} \oplus \operatorname{range} T^{\dim V - 1}. \qquad \square$$

20 Suppose V is an inner product space and $T \in \mathcal{L}(V)$ is normal and nilpotent. Prove that $T = 0$.

Solution:

Consider the operator T^*T, which is positive (Theorem 7.64), and hence self-adjoint. By the Spectral Theorem (either complex or real), T^*T is diagonalizable. If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T^*T, then the eigenvalues of $(T^*T)^k$, where k is some integer, are $\lambda_1^k, \dots, \lambda_n^k$ (which follows from *Problem 5C.2*).

Now examine $(T^*T)^{\dim V}$

$$(T^*T)^{\dim V} = (T^*)^{\dim V} T^{\dim V} = 0,$$

as $T^{\dim V} = 0$ (by Theorem 8.16). So, the only eigenvalue of $(T^*T)^{\dim V}$ is zero, and therefore the only eigenvalue of T^*T is zero, hence $T^*T = 0$, which follows from diagonalization of T^*T. Now the *Problem 7A.2* implies $T = 0$, as desired. $\square$

21 Suppose $T \in \mathcal{L}(V)$ is such that $\operatorname{null} T^{\dim V - 1} \neq \operatorname{null} T^{\dim V}$. Prove that T is nilpotent and that $\dim \operatorname{null} T^k = k$ for every integer k with $0 \leq k \leq \dim V$.

Solution:

First, $\operatorname{null} T^{\dim V - 1} \neq \operatorname{null} T^{\dim V}$ implies the dimension of $\operatorname{null} T^k$ decreases by at least one going from k to $k-1$, otherwise Theorem 8.2 would imply $\operatorname{null} T^{\dim V - 1} = \operatorname{null} T^{\dim V}$.

Now note $\dim \operatorname{null} T^{\dim V} \leq \dim V$, which follows from the Fundamental Theorem of Linear Maps. Using the conclusion of the previous paragraph, we have

$$\dim \operatorname{null} T \leq \dim V - (\dim V - 1) = 1.$$

Thus, $\dim \operatorname{null} T$ equals either 1 or 0. If $\dim \operatorname{null} T = 0$, then $\operatorname{null} T = 0$ and Theorem 8.2 would imply $\operatorname{null} T^0 = \operatorname{null} T = \operatorname{null} T^{\dim V - 1} = \operatorname{null} T^{\dim V}$, contradicting one of the initial assumptions.

Now an increase in the power of T^k by one should lead to an increase in dimension of the null space by one too. If the dimension increases by more than 1, the $\operatorname{null} T^k$ becomes V at some $k < \dim V$, which would imply once again $\operatorname{null} T^{\dim V - 1} = \operatorname{null} T^{\dim V}$. That shows $\dim \operatorname{null} T^k = k$ for every integer k with $0 \leq k \leq \dim V$

Finally, as $\dim \operatorname{null} T^{\dim V} = \dim V$, we have $\operatorname{null} T^{\dim V} = V$, hence $T^{\dim V} = 0$, and therefore T is nilpotent, by definition. $\square$

22 Suppose $T \in \mathcal{L}(\mathbb{C}^5)$ is such that $\operatorname{range} T^4 \neq \operatorname{range} T^5$. Prove that T is nilpotent.

Solution:

If $\operatorname{range} T^4 \neq \operatorname{range} T^5$, *Problem 8A.6* implies $\dim \operatorname{range} T^4 > \dim \operatorname{range} T^5$. Using the Fundamental Theorem of Linear Maps, we can transform it into $\dim \operatorname{null} T^4 < \dim \operatorname{null} T^5$. Now the result of *Problem 8A.21* implies that T is nilpotent. $\square$

23 Give an example of an operator T on a finite-dimensional real vector space such that 0 is the only eigenvalue of T but T is not nilpotent.

Solution:

Let $V = \mathbb{R}^3$. Define T by $T(x, y, z) = (y, -x, 0)$. T has the following matrix with respect to the standard basis:

$$\mathcal{M}(T) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Its only eigenvalue is zero, but it is not nilpotent. Using the matrix multiplication it is easy to show that

$$\mathcal{M}(T^3) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

If T were nilpotent, we would have had $T^3 = 0$, following Theorem 8.16. $\square$

24 For each item in Example 8.15, find a basis of the domain vector space such that the matrix of the nilpotent operator with respect to that basis has the upper-triangular form promised by 8.18(c).

Solution:

(a) The operator $T \in \mathcal{L}(\mathbb{F}^4)$ defined by

$$T(z_1, z_2, z_3, z_4) = (0, 0, z_1, z_2).$$

Let $e_1 = (0, 0, 1, 0)$, $e_2 = (0, 0, 0, 1)$, $e_3 = (1, 0, 0, 0)$, $e_4 = (0, 1, 0, 0)$. With respect to this basis, T has the matrix

$$\mathcal{M}(T) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

which is upper-triangular, as desired.

(b) The operator T on $\mathbb{F}^3$ whose matrix (with respect to the standard basis) is

$$\begin{pmatrix} -3 & 9 & 0 \\ -7 & 9 & 6 \\ 4 & 0 & -6 \end{pmatrix}$$

Let $v_1 = (3, 1, 2)$, $v_2 = (1, 1, 0)$, $v_3 = (0, 0, 1)$. With this basis

$$\begin{aligned} Tv_1 &= 0, \\ Tv_2 &= 2v_1, \\ Tv_3 &= -2v_1 + 2v_2. \end{aligned}$$

Hence, the matrix of T with respect to basis v_1, v_2, v_3 is

$$\begin{pmatrix} 0 & 2 & -2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

(c) The operator of differentiation on $\mathcal{P}_m(\mathbb{R})$. Take the basis $1, x, x^2, \ldots, x^m$. The matrix of D with respect to it is

$$\begin{pmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 2 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & m \\ 0 & 0 & 0 & \ldots & 0 \end{pmatrix},$$

which is upper-triangular.

25 Suppose that V is an inner product space and $T \in \mathcal{L}(V)$ is nilpotent. Show that there is an orthonormal basis of V with respect to which the matrix of T has the upper-triangular form promised by 8.18(c).

Solution:

By Theorem 8.18, the minimal polynomial of T is z^m for some positive integer m. Hence, Theorem 6.37 implies that T has an upper-triangular matrix with respect to some orthonormal basis of V. Theorem 8.17 states that 0 is the only eigenvalue of T, hence all diagonal entries of the matrix of T are zero, so it is the matrix of the needed form. $\square$